

Rui-Yang Zhang

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EDUCATION

Lancaster University

PhD Statistics and Operation Research

Lancaster, UK

October 2024 - Current

- **Project Title:** Bayesian sequential data acquisitions for ocean engineering and probabilistic numerics
- **Project Partner:** ARC TIDE Research Hub
- **Supervisors:** David Leslie, Henry Moss, Lachlan Astfalck, Edward Cripps

Lancaster University

MRes Statistics and Operation Research

Lancaster, UK

October 2023 - September 2024

- **Grade:** Distinction

University College London

BSc Mathematics and Statistical Science

London, UK

September 2020 - June 2023

- **Grade:** First Class (Dean's List & RSS Award)

PREPRINTS AND PUBLICATIONS

1. Iguchi, Y., Livingstone, S., Vasdekis, G., & **Zhang, R. Y.** (2026). Pathwise skew-symmetric discretisation for SDEs with superlinear drift. *arXiv: 2607.17735*.
2. **Zhang, R. Y.**, Astfalck, L. C., Cripps, E. J., and Leslie, D. S., Moss, H. B. (2026). Dynamic Gaussian Processes and the Vanilla-SPDE Exchange. *arXiv: 2606.31063*.
3. **Zhang, R. Y.**, Astfalck, L. C., Cripps, E. J., and Leslie, D. S., Moss, H. B. (2026). BALLAST: Bayesian Active Learning with Look-ahead Amendment for Sea-drifter Trajectories under Spatio-Temporal Vector Fields. **Proceedings of the 43rd International Conference on Machine Learning**.
4. Iguchi, Y., Livingstone, S., Nüsken, N., Vasdekis, G., & **Zhang, R. Y.** (2026). Skew-symmetric schemes for stochastic differential equations with non-Lipschitz drift: an unadjusted Barker algorithm. **IMA Journal of Numerical Analysis** (forthcoming).
5. **Zhang, R. Y.**, Moss, H. B., Astfalck, L. C., Cripps, E. J., and Leslie, D. S. (2024). BALLAST: Bayesian Active Learning with Look-ahead Amendment for Sea-drifter Trajectories. **NeurIPS 2024 Workshop on Bayesian Decision-making and Uncertainty**.

TALKS

- Dynamic Gaussian Processes and the Vanilla-SPDE Exchange.
 - Non-Equilibrium Sampling Workshop, **Newcastle University**. September 2026.
 - MCQMC 2026, **University of Edinburgh**. June 2026.
 - CSML Reading Group, **Lancaster University**. October 2025.
 - Workshop on Kernels, Flows, and Sampling, **Newcastle University**. October 2025.
- BALLAST: Bayesian Active Learning with Look-ahead Amendment for Sea-drifter Trajectories under Spatio-Temporal Vector Fields
 - Newcastle Departmental Seminar, **Newcastle University**. October 2026.
 - MARS Annual Symposium, **Lancaster University**. September 2026.
 - DTU Departmental Seminar, **Technical University of Denmark**. June 2026.
 - AI Across Scales Workshop, **Isaac Newton Institute**. February 2026.
 - Irish CRT Winter Symposium, **University College Dublin**. January 2026.

- STOR-i Annual Conference, **Lancaster University**. January 2026.
- STOR-i Forum, **Lancaster University**. October 2025.
- Physics Enhancing Machine Learning Workshop 2025, **Institute of Physics**. October 2025.
- BALLAST: Bayesian Active Learning with Look-ahead Amendment for Sea-drifter Trajectories
 - Statistics PhD Seminar, **University College London**. February 2025.
 - Biannual TIDE Workshop, **University of Western Australia**. November 2024.
 - STOR-i Forum, **Lancaster University**. October 2024.
- Skew-symmetric schemes for stochastic differential equations with non-Lipschitz drift
 - STOR-i Forum, **Lancaster University**. June 2024.
 - CSML Reading Group, **Lancaster University**. June 2024.

GRANTS AND AWARDS

- Early Career Researcher Travel Grant for MCQMC 2026, **ProbAI Hub**, *June 2026*.
- Visiting Fellowship in Machine Learning with Chris Oates, **Leverhulme Trust**, *April 2026 - April 2027*.
- Early Career Researcher Travel Grant for Physics Enhancing Machine Learning 2025, **Data-Centric Engineering**, *October 2025*.
- PGR Citizenship Prize (with A. Makris), **School of Mathematical Sciences, Lancaster University**, *July 2025*
- Focused Research Grants (with Y. Iguchi, S. Livingstone, G. Vasdekis), **Heilbronn Institute for Mathematical Research**, *June 2025*
- Funding for Academic Visit and Workshop Presentation, **ARC TIDE Research Hub**, *November 2024*.
- Early Career Researcher Travel Grant, **London Meeting on Computational Statistics**, *June 2024*.
- Royal Statistical Society Award, **UCL**, *September 2023*.
- Dean's List, **UCL**, *September 2023*.
- LMS Undergraduate Research Bursary, **London Mathematical Society**, *June 2023 - July 2023*.

SUPERVISIONS

Benjamin Matsuert <i>Probabilistic Eddy Identification using Gaussian Processes</i>	STOR-i Summer Internship <i>July 2025 - August 2025</i>
Harriett Du Four (Secondary Supervisor) <i>Spatial Analysis of Sea Surface Temperature</i>	STOR-i Summer Internship <i>July 2025 - August 2025</i>

ACTIVITIES AND SERVICE

Research Visit to Chris Oates' Group <i>Visitor</i>	Newcastle, UK <i>October 2026 - November 2026</i>
Research Visit to Søren Hauberg's Group <i>Visitor</i>	Copenhagen, DK <i>May 2026 - June 2026</i>
Learning Development Maths and Stats Hub <i>Tutor</i>	Lancaster, UK <i>October 2025 - July 2026</i>
STOR-i Forum <i>Co-Organiser</i>	Lancaster, UK <i>October 2024 - September 2026</i>
Outreach Talks to Sixth Form Students <i>Speaker</i>	Lancaster, UK <i>March 2025 - April 2025</i>
Lancaster AI Reading Group <i>Co-Organiser</i>	Lancaster, UK <i>October 2024 - September 2025</i>

Research Visit to UCL

Visitor

London, UK

June 2025

Research Visit to TIDE Hub at UWA

Visitor

Perth, AU

November 2024

BayesAI Workshop

Co-Organiser

Lancaster, UK

June 2024 - September 2024

SKILLS

Languages: English (Bilingual), Mandarin Chinese (Native)

Programming Languages: R, Python (JAX, Numpy, GPJax)